

CHAPTER TEN

VERTICAL PRICE RELATIONS IN AGRICULTURE

STEPHAN VON CRAMON-TAUBADEL

10.1. Introduction and learning objectives

With few exceptions, consumers in industrialised countries buy little food directly from the farm. Instead, most agricultural products are consumed at the end of a long and multi-linked chain from farm to fork. Along this chain, agricultural products are sorted, inspected, transported, stored, packaged and placed on retail shelves. In most cases they are also physically transformed and combined with other products. For example, apples become juice, cattle become cuts of meat, sausage and leather, and tomatoes, onions and broccoli are combined with other ingredients to make vegetarian pizza.

Hence, the money that consumers spend on food does not only pay for the agricultural raw products contained in this food, but also for the many goods and services that are added to these raw products between the farm and the fork. In 2018, farmers in Germany for example received on average 20.8 cents of every euro consumers spent on food (see Figure 10.1). However, this share varies considerably across products and time. First, while grain farmers received only 4.1% of consumer expenditure for bread in 2018, egg producers received 48.5% of consumer expenditure on eggs in shell. Second, the farm share of consumer expenditure on food has fallen over time. In the early 1970s, farmers in Germany received 47.5 cents of every euro (or DM) that consumers spent on food, more than twice as much as 20.8 cents in 2018. However, Figure 10.1 also shows that the farm share of

consumer expenditure on food has stabilised over the last roughly 10 years and may even be rising again for eggs.¹

Figure 10.1: The farmers' share of consumer expenditure on food in Germany (%)



Source: Thünen Institute (2020), own depiction. Data for West Germany up to and including 1990/91. After the 2004/05 marketing year, reporting changed to calendar years.

The farm share of the consumer food bill is a controversial topic. It is often argued that farmers are subject to unfair competition and do not receive a fair share of consumer expenditure on food because of the structure of competition in the food chain (with often only few processors and retailers dominating the chain between farmers and consumers). A typical accusation is that food processors pass increases in the prices of raw agricultural products on to consumers faster and more completely than reductions in raw material prices. Sometimes consumers are accused of inappropriate behaviour as well. It is argued that bargain-hunting consumers do not show sufficient appreciation for the true value of food, and that the resulting pressure on prices is passed back along the food chain to farmers. This has led to calls for policy makers to enforce minimum retail prices to ensure that food is not sold below its cost of production. In Germany in recent years,

¹ Producing data on the farmers' share of consumer expenditure on food is a complex undertaking. Sinabell (2005) discusses the statistical challenges involved. Schnepf (2015) presents facts and figures on the development of the farmers' share of consumer expenditure on food in the USA.

this discourse has triggered several so-called food summits and other initiatives involving high-ranking politicians and representatives of agriculture, the processing industry, retailers and consumers.²

Against the background of these sometimes very heated discussions, what can economic analysis tell us about so-called **vertical price relationships**³ in the food chain? Some of the issues raised above, such as the definition of the intrinsic value of food, are beyond the scope of economic analysis. However, economics can formulate models that provide insights into how prices in vertically integrated markets react to one another under certain conditions, what factors influence these price reactions, and what consequences they have for the actors in the food chain. The transmission of price signals along food and other chains is crucial for the efficient functioning of a market economy. Therefore, in this chapter we provide an introduction into the economic analysis of the relationships between the prices of agricultural raw products at the farm gate and the prices of corresponding food products along the food chain, e.g. at the wholesale and retail levels.

In this chapter we:

- discuss which factors influence vertical price relations and, consequently, the farm share of consumer expenditure on food;
- describe how these factors have changed over time and why farmers in Germany received 47.5 cents of every euro spent by consumers on food in the early 1970s, but only 20.8 cents in 2018;
- outline the models and empirical methods that economists use to analyse vertical price relations in agricultural and food markets;
- analyse the factors that determine the speed with which price changes at one stage in the food chain are passed on to other stages;

² For example, one such summit took place on 3 February 2020 in the Federal Chancellery in Berlin: see

<https://www.agrarheute.com/politik/live-merkel-kloeckner-altmaier-treffen-lebensmittelwirtschaft-564454>.

³ Price relationships along the chain from farm to fork are often referred to as vertical price relationships to distinguish them from spatial price relationships, which are sometimes referred to as spatial.

- discuss whether and how concentration in many branches of food processing affects vertical price formation (i.e. whether processing or retail companies use their market power to distort vertical price relations to the disadvantage of farmers and/or consumers); and
- explain what is meant by the term asymmetric price transmission, and what can cause asymmetric price transmission in the food chain.

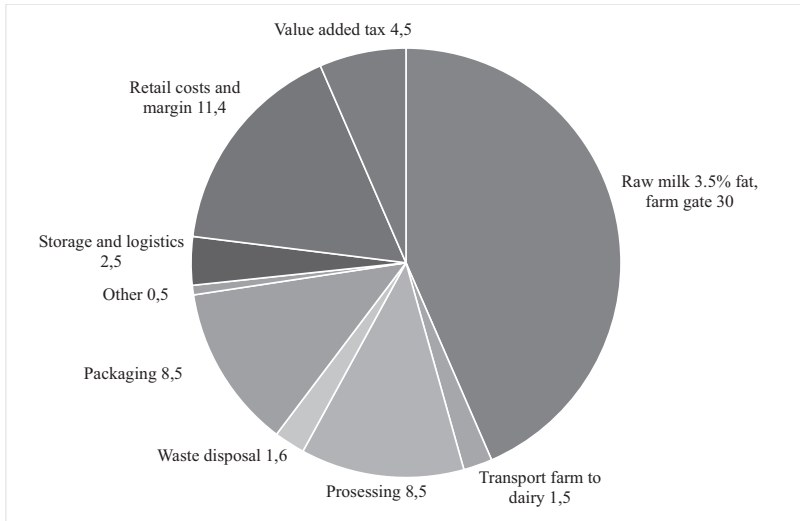
10.2. The determinants of the long-term evolution of the farm-retail price spread in agriculture

We begin by considering why the farm share of consumer expenditure on food has tended to fall over time, as shown for the example of Germany in Figure 10.1. Consumer expenditure on a food product is the sum of payments to farmers for the agricultural raw products that they produce, and payments to the suppliers of the **complementary goods and services** (CGS) that are added to these raw products along the chain. The following formula illustrates this relationship.

$$P_{Cons} = P_{raw}\alpha_{raw} + \sum_{i=1}^k P_i\alpha_i \quad (1)$$

where P_{cons} denotes the per unit consumer price of the final food product, and P_{raw} the per unit price of the agricultural raw product. α_{raw} is a **transformation coefficient** that indicates how many units of the raw product are needed to produce one unit of the consumption product. The farm share of consumer expenditure on the food product is thus the ratio $\frac{P_{raw}\alpha_{raw}}{P_{cons}}$. The **unit margin** $\sum_{i=1}^k P_i\alpha_i$ is the total value of the $i = 1, 2, \dots, k$ CGS that are added to the agricultural raw product; P_i is the per unit price of the i -th CGS, and α_i as the corresponding transformation coefficient. Figure 10.2 illustrates these relationships for the example of fluid milk in Germany in October 2018. The consumer price of milk was 69 cents/litre, of which the farm-gate price of raw milk accounted for 30 cents/litre, and the various CGS added between the farm and retail for the remaining 39 cents/ litre.

Figure 2: The composition of the price fluid milk in Germany (October 2018, in cents/litre)



Source: Own depiction with data from Institut für Ernährungswirtschaft (2018).

If P_{cons} in equation (1) is multiplied by the total quantity demanded of the final food product Q_{cons} , we obtain

$$P_{cons} Q_{cons} = P_{raw} \alpha_{raw} Q_{cons} + \sum_{i=1}^k P_i \alpha_i Q_{cons} \quad (2)$$

or

$$consumer\ expenditure = P_{raw} Q_{raw} + \sum_{i=1}^k P_i Q_i \quad (3)$$

since $\alpha_{raw} Q_{cons}$ equals the quantity of the raw agricultural product (Q_{raw}) that is required to produce Q_{cons} , and $\alpha_i Q_{cons}$, by analogy, equals the required quantity of the i -th CGS (Q_i).

The sum $\sum_{i=1}^k P_i Q_i$ in equation (3) is called the **market margin**. We see that the market margin is the product of a **price vector** $\{P_1, P_2, \dots, P_k\}$ and a **quantity vector** $\{Q_1, Q_2, \dots, Q_k\}$. The determinants of the market margin can thus be divided into those that influence the quantity vector, and those that influence the price vector (Koester 2016, p. 142 ff.).

10.2.1. Factors that influence the quantity vector

- a) **Rising incomes.** The demand for most food is known to be income inelastic in industrialised countries such as Germany. The demand for CGS is usually more income elastic. The demand for CGS therefore expands more as a result of an income increase than the demand for raw agricultural products. For example, we do not necessarily demand more potatoes as our incomes rise, but we purchase more processed potato products and eat in restaurants more often, where we pay to have potatoes prepared into different dishes. Hence, as incomes rise, the quantities of CGS (Q_i) in equation (3) increase and/or the dimension of the quantity vector (k) increases as new CGS are added to a food product. Both of these changes (more CGS and new CGS) increase the market margin.
- b) **Household structures and preferences.** Economic development and rising incomes have been accompanied by far-reaching changes in the structure and preferences of households, not only in industrialised countries. Women, who in many societies traditionally prepared meals and thus added many CGS to food (usually without pay), are increasingly employed outside the home and spend less time preparing meals. As women's opportunity costs have increased, so has the willingness of households to pay for so-called **convenience foods** that are more processed and easier to prepare. This leads to an increase in the dimension k in equation (3): CGS that used to be added in households (free of charge) are increasingly added by firms in the food chain (and paid for).
- c) **The outsourcing of CGS formerly added on farms.** Not only consumer households but also farms have outsourced food processing functions over time. Many farms used to, for example, process milk into butter and cheese, or slaughter animals and produce sausages and other meat products for sale. As farms have become more specialised, firms in the food chain have taken over these processing steps. In terms of equation (3), this means that in the past the raw product sold by the farmer was often partially processed and thus less 'raw' than it is today. Hence, over time some CGS have shifted from farm income to the market margin.

The first two factors listed above, rising incomes and the outsourcing of processing steps from consumer households, both increase the market

margin but, *ceteris paribus*, leave farm revenues largely unchanged.⁴ Therefore, the farm share of consumer expenditures for food falls, but farm incomes are not negatively affected. However, if farms outsource functions and sell products that are less processed over time (the third factor listed above), then farm-gate prices and farm revenues fall. This does not necessarily imply that farm household welfare will fall. On-farm processing steps requires labour and other resources that are saved when these functions are outsourced. Farm household members are freed to pursue other, more rewarding work, or to enjoy more leisure time. Furthermore, specialisation in pure agricultural production can lead to efficiency gains and better utilisation of economies of scale, thus increasing the overall profitability of the farm.

10.2.2. Factors that influence the price vector

Changes in the price vector can also affect the market margin. Declining **agricultural terms of trade** compared with other sectors over many decades means that the prices of agricultural products have tended to fall relative to the prices of many industrial products and services. If P_{raw} falls relative to an index of the CGS prices $\{P_1, P_2, \dots, P_k\}$ in equation (3), the market margin will increase. State intervention can also play a role, for example when the prices of agricultural commodities are supported, as was (and is) the case in many industrialised countries. For example, the steady reduction of price support for agricultural products in the EU that began with the so-called MacSharry-reform of 1993, had tended to reduce P_{raw} and, thus, increase the market margin.⁵

Taxes and duties can also contribute to a widening of the market margin. In the example of fluid milk presented in Figure 10.2, value added tax accounts for 4.5 cents/litre or roughly 11.5% of the market margin. A further 1.6 cents/litre (4.1% of the market margin) is attributable to waste disposal costs under the so-called Grüner Punkt system in Germany.⁶ For some consumer products of agricultural origin (e.g. cigarettes), taxes account for over half of the consumer price. Government regulations also influence the market margin, for example best-before dates, nutrition labelling, minimum wages for workers in the food processing industry, and the increased

⁴ Rising incomes will lead to some increase in the demand for food, and thus boost farm revenues, depending on income elasticities for different types of food.

⁵ EU price support via the intervention system and its reduction over time is discussed in Chapter 9.

⁶ See <https://www.gruener-punkt.de/en/company/about-us>.

documentation and labelling that is required to ensure the traceability of food. These regulations both add CGS to the quantity vector of the market margin and increase the prices of individual CGS in the market margin.

The factors discussed above have contributed to a widening of the market margin over time. As shown Figure 10.1, the farm share of consumer expenditure on food in Germany has tended to fall and, conversely, the market margin has grown. The situation in many other countries is similar. However, we should be cautious about extrapolating these trends into the future. As noted above, there are signs in Figure 10.1 that the farm share of consumer expenditure on food has stabilised over the last decade in Germany. This has been helped by the fact that the trend of falling real agricultural prices has slowed down and, at least in phases (notably during the so-called agricultural price crisis of 2007/08), has been reversed. The boom in cooking at home and the popularity of cooking shows and celebrity chefs has led to many households buying less processed food and preparing it more elaborately themselves; the number of farms selling agricultural products directly to consumers (farm shops, weekly markets, delivery services) has increased.

Whether these trends will continue remains to be seen. The Covid-19 pandemic will likely lead to at least a temporary decline in the market margin because of the associated reductions in household incomes, but also because eating away from home (e.g. in restaurants and canteens) has fallen sharply due to infection control restrictions. The pandemic has led to a scarcity of seasonal workers in many countries and, thus, to reductions in supply and increases in the farm-gate prices of fruit and vegetables. This will likely reduce market margins for these products, but other pandemic-related factors such as transportation and logistic bottlenecks will work in the opposite direction. In the longer term, changes in the structure and functions of trade (e.g. new platforms such as online trade or online retail shopping) will also affect the level and evolution of the market margin.

10.3. Gardner's model

The previous section discussed factors that influence the market margin between farm-gate and consumer prices for food in the long run. These factors include rising incomes and changes in both farm and consumer preferences. In the long run, these factors have led to the addition of more and more CGS in the food chain. In the following section, we assume that incomes and preferences are given and examine the short-run relationships

between farm-gate and consumer prices - the so-called vertical price relationships. The aim is to better understand the factors that influence whether and how fully consumer prices and producer prices react to one another.

10.3.1. Basics

The starting point for much of the literature on vertical price relationships in recent decades is a model that was originally proposed by Gardner (1975). The title of Gardner's seminal paper refers to the *farm-retail price spread*, a term that is often used to describe the margin between farm-gate and consumer prices. In Gardner's model, under conditions of perfect competition, food-processing firms (the processors) produce a final consumer product x using two inputs; a raw agricultural product a and a non-agricultural input (or bundle of CGS) b . The first equation in the model is a linear homogeneous production function⁷ that describes how a and b are transformed into the consumer product x :

$$x = f(a, b) \quad (\text{supply of the consumer product } x) \quad (4)$$

The production function in equation (4) is not necessarily limitational, i.e. a and b do not have to be combined in fixed proportions. In equations (1) to (3) above we implicitly assumed limitational production technology defined by the fixed transformation coefficients α_{raw} and α_i . Instead, the production function in equation (4) allows for substitution between inputs depending on the **elasticity of factor substitution** σ . σ can be interpreted as a measure of the curvature of the isoquants of a production function and is defined as:

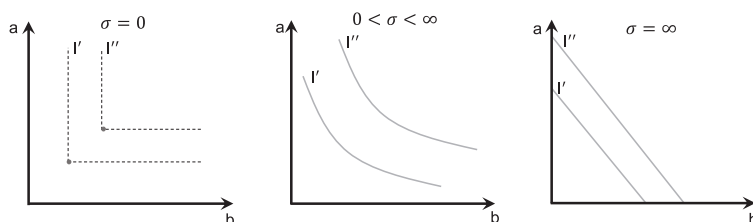
$$\sigma = \frac{\partial(a/b) / (a/b)}{\partial(P_b/P_a) / (P_b/P_a)} \quad (5)$$

Equation (5) shows that the elasticity of substitution indicates which proportional adjustment of the factor input ratio (a / b) is triggered by a 1% change in the factor price ratio (P_b/P_a). If the production technology is limitational, even large changes in the factor price ratio do not lead to an adjustment in the factor input ratio. This is because input use is governed by fixed transformation coefficients and more of both inputs, in fixed proportion, is needed to increase production. In this case the elasticity of

⁷ See Appendix 2.3 in Chapter 2.

factor substitution equals 0 and the isoquants are shaped as right angles (Figure 10.3, left panel). At the other extreme, if the isoquants are straight lines, a very small change in the factor price ratio leads to a jump from one corner solution with $(a/b) = 0$ to the other corner solution with $(a/b) = \infty$ (or vice versa). In this case the elasticity of factor substitution equals ∞ (Figure 10.3, right panel). If $0 < \sigma < \infty$, then the isoquants are the familiar convex (from the origin) curves (Figure 10.3, middle panel) that are typically shown in textbooks. In this case some, but not unlimited input substitution is possible.

Figure 10.3: Different isoquants and elasticities of factor substitution



Source: Own representation.

The elasticity of factor substitution σ depends on the production technology and is often relatively low in food processing technologies. For example, the production of potato chips (x) requires a certain quantity of potatoes (a); these cannot be substituted in the short term, or only to a very limited extent, by using more CGS such as labour, energy and capital in the input bundle (b). However, technical progress can increase σ in the medium and long term. For example, breeders might develop new potato varieties whose shapes allow a higher yield of potato chips.

In addition, the elasticity of factor substitution can also be influenced by institutions. One example is the German *Reinheitsgebot* (purity law) for beer, which prohibits input substitutions that are technically possible and permitted in other countries (e.g. brewing beer with maize or rice rather than barley).⁸ Another example are standards for the use of natural and other flavours (case study 1).

⁸ The *Reinheitsgebot* stipulates that, with a few exceptions, beer can only be brewed with four ingredients: hops, barley, yeast and water. The example of the *Reinheitsgebot* shows that laws that restrict substitution possibilities in the

Case study 1: Standards and substitution possibilities in the production of raspberry yoghurt in Germany

Producers can add raspberry flavour to 100 kg of yoghurt in a variety of ways (Ahlers 2019). First, fresh raspberries can be used, which costs about 31.50 euros. Natural raspberry flavouring extracted from raspberries costs about 12.50 euros. Raspberry flavouring, which can be obtained from fungal cultures and sawdust, for example, costs about 3.75 euros. Finally, there are raspberry flavouring preparations that can be produced from artificial raw materials and that cost only about 0.06 euros. These various technical possibilities give producers considerable scope (and considerable financial incentives) to substitute the agricultural raw product raspberry with other, non-agricultural inputs. However, labelling regulations limit these substitution possibilities. If the words "natural raspberry flavouring" appears on the label, then no less than 95% of the flavouring must have been obtained from either fresh raspberries or natural raspberry flavouring (Verbraucherzentrale 2016). If raspberry flavouring (for example from fungal cultures) was used, then the words "natural flavouring" (but not "natural raspberry flavouring") may appear on the label. Consumer advocate groups in Germany complain that this is misleading and lobby for stricter labelling requirements that would increase transparency for consumers but also further limit substitution possibilities in food processing.

The production function in equation (4) describes the supply of the consumption product x as a function of the input quantities a and b . Gardner's model is completed by five further equations that describe the demand for x as well as supply and demand for a and b .

$$x = g(P_x) \quad (\text{Demand for the consumer product } x) \quad (6)$$

$$a = h(P_a) \quad (\text{Supply of the raw agricultural product } a) \quad (7)$$

production of food are sometimes introduced or maintained for protectionist reasons. For a long time, beer that had been brewed abroad but not in accordance with the *Reinheitsgebot* could not be sold as beer in Germany. The enforcement of the *Reinheitsgebot* thus restricted foreign competition on the German beer market. In 1987, the European Court of Auditors ruled that this violated the free movement of goods in the EU internal market. Since then, foreign beers that are not brewed according to the *Reinheitsgebot* can be offered for sale in Germany as beer (see, for example, Dornbusch 2013).

$$\frac{\partial(f(a, b))}{\partial a} P_x = P_a \quad (\text{Demand for the agricultural raw product } a) \quad (8)$$

$$b = k(P_b) \quad (\text{Supply of the non-agricultural input } b) \quad (9)$$

$$\frac{\partial(f(a, b))}{\partial b} P_x = P_b \quad (\text{Demand for the non-agricultural input } b) \quad (10)$$

Here, g , h and k denote functions that satisfy plausible, general conditions. For example, the function g should be such that the demand for x falls as P_x rises, i.e. $\partial(g(P_x))/\partial P_x < 0$. Equations (8) and (10) describe the derived demand for factors of production a and b under conditions of perfect competition.⁹ The model thus comprises a total of six equations (4, 6, 7, 8, 9 and 10) and six unknowns (the quantities x , a and b , and the corresponding prices P_x , P_a , and P_b).

Comparative-static analyses are performed on the model by first totally differentiating and linearising all equations. These steps are detailed in Appendix 1.¹⁰ Equation (7), for example, gives rise to:

$$\dot{a} = \varepsilon_{a, P_a} \dot{P}_a \quad (7')$$

with $\dot{a} = da/a$, $\dot{P}_a = dP_a/P_a$ and ε_{a, P_a} the own-price elasticity of supply of a .

This is often called the **equilibrium displacement form of an equation** because it can be used to study transitions between equilibrium states that are triggered by shocks. It must be emphasised that equations like (7') are only local approximations. They provide good approximations of the behaviour of the variables in question in response to small changes (say 5% or 10%). Hence, equation (7') tells us that in response to a small percentage change in P_a (measured by $\dot{P}_a$), a will change by approximately $\varepsilon_{a, P_a} \dot{P}_a$ percent. However, since we do not know the exact mathematical form of the underlying function (in this case the supply function $a = h(P_a)$), the

⁹ In equations (8) and (10), the marginal value product (marginal product*output price) is on the left-hand side, and the factor price is on the right-hand side. See the explanation of marginal productivity theory in Chapter 2.

¹⁰ These are the same steps used in section 3.4.2 of Chapter 3 (to model the impact of a product price increase on land prices), and in section 4.7 of Chapter 4 (to model the impact of income and technical progress on agricultural prices).

approximation in equation (7') can become increasingly inaccurate for larger changes. Equilibrium displacement models are very useful tools for studying the behaviour of interdependent variables in economics; however, one should be wary of applications in which they are used to simulate the consequences of large changes such as doubling or halving of prices or quantities.

10.3.2. The elasticity of vertical price transmission - the response to producer price changes

The *equilibrium displacement* form of Gardner's model, shown in Appendix 1, can be used to examine the effects of various exogenous shocks. For example, to determine the impact of a change in the price of the raw agricultural product a on the price of the consumer product x , the supply equation for a (equation (7)) is replaced by $\dot{P}_a = \dot{\bar{P}}_a$, where $\dot{\bar{P}}_a$ represents an exogenously given proportional change in the price of a . Such an exogenous change could be triggered, for example, by the state adjusting the support price for a , which is what the EU used to do each year when it fixed the intervention prices for agricultural products. The system is then solved for the so-called **elasticity of vertical price transmission**: $\frac{\dot{P}_x}{\dot{\bar{P}}_a}$ (henceforth E_{P_x, P_a}). The result is:¹¹

$$E_{P_x, P_a} = \frac{S_a(\sigma + \varepsilon_{b, P_b})}{\varepsilon_{b, P_b} + S_a\sigma - S_b\eta_{x, P_x}} \quad (11)$$

where $S_i = \frac{iP_i}{xP_x}$ is the cost share of production factor i , σ is the elasticity of substitution between factors a and b , ε_{i, P_i} is the own-price elasticity of supply of factor i , and η_{x, P_x} is the own-price elasticity of demand for the consumer product x .

All terms in equation (11) are positive except for the own-price elasticity of demand for x (which, however, is in a term with a minus sign). Hence, we can conclude that $E_{P_x, P_a} > 0$. Moreover, it can be proved as follows that $E_{P_x, P_a} < 1$:

$$E_{P_x, P_a} < 1 \text{ if } S_a(\sigma + \varepsilon_{b, P_b}) < \varepsilon_{b, P_b} + S_a\sigma - S_b\eta_{x, P_x}$$

¹¹ Equation (19) in Gardner (1975).

$$S_a\sigma + S_a\varepsilon_{b,P_b} < \varepsilon_{b,P_b} + S_a\sigma - S_b\eta_{x,P_x}$$

$$S_a\varepsilon_{b,P_b} - \varepsilon_{b,P_b} < -S_b\eta_{x,P_x}$$

$$(S_a - 1)\varepsilon_{b,P_b} < -S_b\eta_{x,P_x}$$

$$(1 - S_a)\varepsilon_{b,P_b} > S_b\eta_{x,P_x}$$

$$\varepsilon_{b,P_b} > \eta_{x,P_x}$$

since under full competition $P_a a + P_b b = P_x x$ and thus $S_a + S_b = 1$ or $S_b = 1 - S_a$.¹²

$\varepsilon_{b,P_b} > \eta_{x,P_x}$ will always be true under normal conditions ($\varepsilon_{b,P_b} > 0$ and $\eta_{x,P_x} < 0$). Thus, $0 < E_{P_x,P_a} < 1$. In other words, a 1% increase in the price of the agricultural commodity a leads to a positive but less than 1% increase in the price of the processed consumer product x .

This result sounds plausible at first. If car tyres increase in price by X%, the price of a new car will increase by less than X%; if tomato prices increase by Y%, the price of a pizza will increase by less than Y%.¹³ The proof presented above can be slightly modified to show that $E_{P_x,P_a} = 1$ will only hold if $\varepsilon_{b,P_b} = \eta_{x,P_x}$, which will never be the case as long as the demand for x falls as the price P_x rises, and the quantity of b supplied increases as the price P_b rises.

Equation (11) can be simplified under certain conditions. If the elasticity of factor substitution is very low ($\sigma \rightarrow 0$), we obtain:

$$E_{P_x,P_a} = \frac{S_a(\sigma + \varepsilon_{b,P_b})}{\varepsilon_{b,P_b} + S_a\sigma - S_b\eta_{x,P_x}} = \frac{S_a(0 + \varepsilon_{b,P_b})}{\varepsilon_{b,P_b} + S_a0 - S_b\eta_{x,P_x}} = \frac{S_a\varepsilon_{b,P_b}}{\varepsilon_{b,P_b} - S_b\eta_{x,P_x}} \quad (12)$$

¹² See equation (81) in Appendix 2.3 in Chapter 2. In a linear homogeneous production function, if all factors are remunerated according to their marginal value products, there is no residual income.

¹³ However, as Kinnucan and Zhang (2015) show, even respected agricultural economists have argued that the elasticity of vertical price transmission should ideally be one (*perfect vertical price transmission*). And it is not uncommon for farmers or consumers to claim instinctively that if slaughter pigs become 5% cheaper, the retail prices of pork chops and bacon should also fall by 5%.

If in addition S_a (the cost share of the raw agricultural product in the consumer product) is relatively large, then since $S_a + S_b = 1$, S_b must be relatively small. Finally, if the demand for the consumer product is inelastic, then the product $S_b \eta_{x, P_x}$ in the denominator of equation (12) will be very small, and we can conclude:

$$E_{P_x, P_a} \approx \frac{S_a \varepsilon_{b, P_b}}{\varepsilon_{b, P_b}} = S_a \quad (13)$$

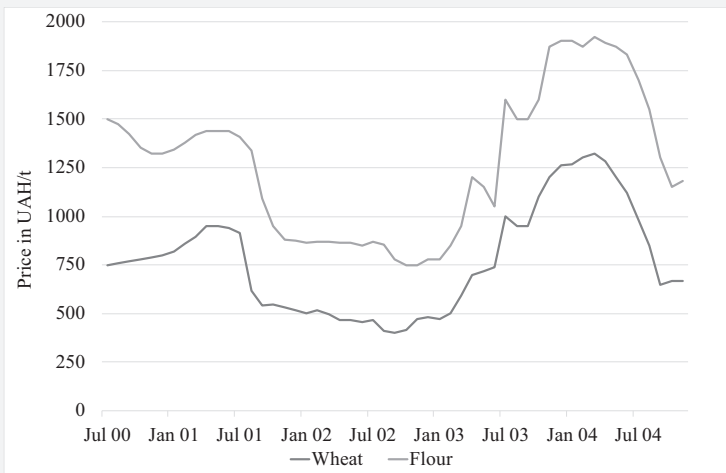
Flour milling is an example of food processing in which these conditions apply. It is possible to produce roughly 0.75 tons of flour from one ton of wheat. There is very little scope for substituting wheat with labour, capital or other inputs in the milling process, at least in the short run.¹⁴ Furthermore, the cost share of wheat in the value of flour is relatively high, and the price elasticity of demand for flour, which is a staple food flour, is low. Case study 2 presents the empirical example of wheat-flour price transmission in Ukraine.

Case study 2: The elasticity of vertical price transmission between wheat and flour in Ukraine

Brümmer et al. (2009) study weekly Ukrainian wheat and flour prices from July 2000 to November 2004. In Figure 10.4 we can see that these prices appear to move together. The large price swings in Figure 10.4 are mainly due to the fact that Ukraine imported wheat following very poor harvests in 2000 and 2003. As a result, import parity prices rather than export parity prices for wheat prevailed in those years.

¹⁴ In the long run, milling machines that yield more flour per ton of wheat could perhaps be developed, but milling technology is mature.

Figure 10.4: Weekly milling wheat and flour prices in Ukraine (July 2000 to November 2004, in Ukrainian Hryvnia (UAH) per ton)



Source: Bruemmer et al. (2009).

The average wheat price over the observation period was 766 Ukrainian Hryvnia (UAH)/t; the average flour price was 1275 UAH/t. Since about 0.75 tons of flour can be obtained from one ton of wheat, 1.33 tons of wheat are needed to produce one ton of flour. These 1.33 tons of wheat cost an average of 1021 UAH ($= 1.33 \text{ t} * 766 \text{ UAH/t}$) during the observation period; the cost share of wheat in flour production was therefore 0.80 ($= 1021 \text{ UAH}/1275 \text{ UAH}$). Thus, in this case all of the conditions for the simplified result $E_{P_x, P_a} \approx S_a$ in equation (13) apply: a low elasticity of substitution σ , a relatively high cost share S_a of the raw agricultural product a , as well as a price-inelastic demand for the consumer product x . Consequently, according to Gardner's model, the elasticity of vertical price transmission between wheat and flour should be roughly equal to $S_a = 0.80$. And indeed, a simple regression between the log wheat and flour prices in Figure 10.4 yields the following results:

$$\ln P_t^{\text{wheat}} = 1.790 + (0.808 * \ln P_t^{\text{wheat}})$$

The estimated elasticity of vertical price transmission between wheat and flour is thus 0.808 (with a standard error 0.015), which is very close to the expected value of 0.80.

10.3.3. The elasticity of vertical price transmission - the response to consumer price changes

In the previous section, the elasticity of vertical price transmission was derived using Gardner's model and the assumption that the price of the agricultural raw product a is exogenous. This elasticity can be referred to as the farm price elasticity of the consumer price and will be labelled E_{P_x, P_a} in the following to illustrate that the farm-gate price P_a has triggered the price adjustments. This elasticity was important in the past when agricultural policy makers in the EU set new intervention prices for agricultural products each year, and economists wanted to forecast the effects of increases or decreases in intervention prices on consumer prices.

However, it is possible to derive another elasticity of vertical price transmission using Gardner's model, specifically E_{P_x, P_a} . This can be called the consumer price elasticity of the farm price, and it measures the effect on the farm-gate price of an exogenous change in the consumer price.¹⁵

$$E_{P_x, P_a} = \frac{\sigma + S_a \varepsilon_{b, P_b} + S_b \varepsilon_{a, P_a}}{\sigma + \varepsilon_{b, P_b}} \quad (14)$$

Since all parameters in equation (14) are greater than zero under normal conditions, E_{P_x, P_a} will always be greater than zero. It can also be shown that $E_{P_x, P_a} < 1$ holds if $\varepsilon_{b, P_b} > \varepsilon_{a, P_a}$. This condition is likely to hold in the vast majority of agricultural applications, especially in an industrial country where food processing is a relatively small sector and acts as a price taker on markets for the CGS in b (e.g. capital, labour and energy).¹⁶ Thus, just as $0 < E_{P_x, P_a} < 1$ is true under normal conditions, so too is $0 < E_{P_x, P_a} < 1$. Yet $E_{P_x, P_a} \neq E_{P_x, P_a}$: the elasticity of vertical price transmission differs

¹⁵ Equation (14) is the inverse of equation (18) in Gardner (1975).

¹⁶ If $\varepsilon_{b, P_b} = \infty$, i.e. the supply of all CGS is infinitely price elastic, then $E_{P_x, P_a} = E_{P_x, P_a} = S_a$, regardless of the values of the other parameters in equations (11) and (14). In the short run, however, it cannot be assumed that the supply of all CGS will be infinitely elastic. Some CGS may be provided by specialised skilled workers or require specialised processing facilities for which the supply curves are upward sloping in the short run.

depending on whether the price adjustments are triggered by a change in the farm-gate price P_a , or a change in the consumer price P_x .

Assume, for example, the following values for the model parameters: $\sigma = 0.5$; $\varepsilon_{a,P_a} = 1$; $\varepsilon_{b,P_b} = 2$; $\eta_{x,P_x} = -0.5$; $S_a = 0.5$. Then $E_{P_x,P_a} = 0.5$, but $E_{P_x,P_a} = 0.8$. In other words, a 1% increase in the producer price leads to a 0.5% increase in the consumer price, but a 0.5% increase in the consumer price leads to an increase in the producer price of only 0.625%. Kinnucan and Zhang (2015, p. 733, Table 1) compare E_{P_x,P_a} and E_{P_x,P_a} for various plausible parameter constellations, and it is always the case that $0 < E_{P_x,P_a} < E_{P_x,P_a} < 1$.

Which elasticity of vertical price transmission should be applied in which situations: E_{P_x,P_a} or E_{P_x,P_a} ? The answer depends on the source of the exogenous shock, in other words which price has triggered the price adjustments. Case study 2 above presents the example of wheat and flour in Ukraine. Since wheat is tradable and Ukraine actively participates in international wheat trade, it is plausible to assume that the price of wheat in Ukraine is largely determined on world markets, i.e. it is exogenous from Ukraine's perspective. Flour, on the other hand, is internationally traded much less than wheat due to its higher perishability and transport costs. Consequently, the flour price in Ukraine is largely determined on a closed domestic market. Therefore, most of the shocks that affect the vertical price relationship between wheat and flour in Ukraine can be expected to arise on the world market for wheat; any shocks that might occur on the Ukrainian flour market will have only negligible effects on international wheat markets. In such a setting, the elasticity E_{P_x,P_a} is appropriate. In contrast, consider price transmission between farm and consumer prices for pork in Germany. Germany exports pork, and some important recent shocks (e.g. the outbreak and spread of African swine fever in China) have presumably affected this export market. These shocks did not arise at the level of the raw product a , but rather at the level of the processed product x . The consequences of such shocks can be better studied with the elasticity E_{P_x,P_a} .

10.4. Vertical price adjustments under imperfect competition

An important assumption of Gardner's model is that of perfect competition. Since we observe high concentration in many branches of the food

processing industry as well as in food retailing in many countries, this assumption can be questioned. Of course, concentration is not a sufficient condition for the exercise of market power. Whether concentration leads to price fixing or other collusive behaviour depends on how firms behave. But as explained in case study 3 in Chapter 8, the German Federal Cartel Office has imposed fines for various forms of collusion in several branches of the food industry over the last 20 years, including milk and dairy products, the coffee market, sausage production, sugar production and food retailing. There are similar concerns about imperfect competition in the food chain in other countries.¹⁷

Various authors have therefore extended Gardner's model to account for market power. These contributions draw on the concept of conjectural variations introduced in Chapter 8. In the case of an oligopoly, the conjectural variation parameter (θ) measures how all other firms together react to a change in quantity by firm i :

$$\theta = \frac{\partial X_{-i}}{\partial x_i} \text{ where } X_{-i} = \sum_{j=1}^n x_j \text{ for } i \neq j \quad (15)$$

θ thus measures how the other firms adjust their supply of the consumer product x when firm i , out of a group of $i = \{1, 2, \dots, n\}$ firms, supplies an additional unit of x . $\theta = -1$ represents aggressive behaviour by competitors; if i lowers its supply by one unit in an attempt to raise the price, this move is fully compensated by the other firms. In this case, the market outcome is identical to the outcome under perfect competition. At the other extreme, $\theta = n - 1$ means that all firms increase or decrease their supply by the same amount, which corresponds to coordinated or collusive behaviour. In this case, the n firms behave together like a monopolist.

In his extension of Gardner's model, Weldegebriel (2004) assumes that processors can exercise oligopoly power over the consumers of x . He therefore adds the parameter θ to the model as a measure of the implementation of oligopoly power on the part of processors. However, Weldegebriel also assumes that processors can exercise oligopsony power vis-à-vis producers of the raw product a . For this purpose, he again resorts to the concept of conjectural variations and defines:

$$\phi = \frac{\partial A_{-i}}{\partial a_i} \text{ with } A_{-i} = \sum_{j=1}^n a_j \text{ for } i \neq j \quad (16)$$

¹⁷ Sexton and Xia (2018), for example, discuss the available evidence for the US.

ϕ measures how the other firms adjust their demand for the raw product a when firm i demands an additional unit of a . Analogous to θ , ϕ lies between -1 and $n - 1$. $\phi = -1$ implies that if firm i demands less of a in an attempt to push the price of a down, the other firms counteract by increasing their demand by an identical amount. $\phi = n - 1$ means that the firms collude and behave together as if they were a monopsonist.

These extensions have consequences for some of the equations in Gardner's model. For example, equation (8), which describes the demand for the raw agricultural product a , becomes:

$$\frac{\partial(f(a,b))}{\partial a} P_x \left(1 + \frac{(1+\theta) x_i}{\eta_{x,P_x}} \right) = P_a \left(1 + \frac{(1+\phi) a_i}{\varepsilon_{a,P_a}} \right) \quad (17)$$

If $\theta = \phi = -1$, then the processing firms behave aggressively towards each other, i.e. the attempts of firm i to exercise market power are always counteracted by the other firms. In this case, equation (17) reduces to equation (8), which holds under perfect competition. If, on the other hand, the n processing firms manage to collude perfectly, then $\theta = \phi = n - 1$. Assuming n identical firms, $\frac{x_i}{x} = \frac{a_i}{a} = \frac{1}{n}$ and equation (17) can be simplified to:

$$\frac{\partial(f(a,b))}{\partial a} P_x \left(1 + \frac{1}{\eta_{x,P_x}} \right) = P_a \left(1 + \frac{1}{\varepsilon_{a,P_a}} \right) \quad (18)$$

Equation (18) describes the factor demand of a firm that is simultaneously a monopolist in the sale of x and a monopsonist in the purchase of a .¹⁸

After the equations in Gardner's model have been modified to account for imperfect competition, the resulting system of equations is solved once more. One important result that we can compare with the case of perfect competition is the farm price elasticity of the consumer price, which now takes the following form:¹⁹

$$E_{P_x, \underline{P_a}} = \frac{(1+\delta)S_a(\sigma + \varepsilon_{b,P_b})}{(1-\mu)(\varepsilon_{b,P_b} + S_a\sigma) - S_b\eta_{x,P_x}} \quad (19)$$

¹⁸ The parameters θ and ϕ are simple mathematical expressions for a variety of possible entrepreneurial strategies, on which there is a very extensive literature. See, for example, Belleflamme and Peitz (2015, chapter 17).

¹⁹ Equation (19) in Weldegebriel (2004).

The parameters δ and μ in equation (19) are functions of the parameters of conjectural variations θ and ϕ . Under conditions of perfect competition, $\delta = \mu = 0$ and equation (19) reduces to equation (11). However, δ and μ also depend on the exact functional forms of the demand for x and the supply of a : the definition of δ contains a term $\frac{\partial \varepsilon_{a,P_a}}{\partial P_a}$ and the definition of μ contains a term $\frac{\partial \eta_{x,P_x}}{\partial P_x}$.²⁰ These terms indicate how the own-price elasticities of the supply of a and the demand for x , respectively, change when the corresponding prices (P_a and P_x , respectively) change. For iso-elastic functions, these terms are zero by definition. For linear and other functional forms, however, this is not the case; the price elasticities vary along the corresponding curves.

Due to this dependence on the exact functional form of the supply and demand curves, no generally valid statements can be made about the effects of market power on vertical price adjustments. Weldegebril (2004) shows that the farm price elasticity of the consumer price E_{P_x, P_a} can be higher but also lower under imperfect competition than under perfect competition. Whether imperfect competition reduces or increases the elasticity depends on the behaviour of the processing firms, and on the functional form of the demand for x and the supply of a . For example, if the functional forms are linear, the supply of a is relatively price elastic ($\varepsilon_{a,P_a} > 1$), and the demand for x is relatively price inelastic ($-1 < \eta_{x,P_x} < 0$), then the exercise of oligopoly power increases the elasticity E_{P_x, P_a} compared with perfect competition. However, when oligopsony power is exercised, E_{P_x, P_a} will be lower. A clear statement on which effect dominates requires a detailed analysis of consumer behaviour (the demand curve $x = g(P_x)$), farmer behaviour (the supply curve $a = h(P_a)$), as well as processor behaviour (the conjectural variations θ and ϕ).

Case study 3: Market power in the Ukrainian milling industry?

Case study 2 presented the following least squares estimation results for the vertical price transmission between wheat and flour prices in Ukraine:

$$\ln P_t^{\text{flour}} = 1,790 + (0,808 * \ln P_t^{\text{wheat}})$$

We noted that the estimated elasticity of vertical price transmission between

²⁰ See equations (13) and (14) in Weldegebril (2004).

wheat and flour of 0.808 is very close to the value of $S_a = 0.80$ that would be expected under conditions of perfect competition according to Gardner's model. Can we conclude from this coincidence that there is perfect competition in the Ukrainian milling industry? Unfortunately, the simple answer to this question is 'no'.

First, note that the milling industry in Ukraine is concentrated: in 2019, according to Latifundist.com (2020), 10 milling firms produced about 47% of flour in Ukraine (0.7674 of 1.630 million tons). Some milling firms are controlled by economically and politically influential so-called oligarchs. Collusive behaviour, e.g. price fixing but also territorial agreements, where competing firms divide the market among themselves into geographical areas, is therefore a distinct possibility. We cannot rule out the possibility that some milling firms simultaneously exercise oligopsony power when they purchase wheat from local farmers and oligopoly power when they sell flour. If this is true, the corresponding effects on vertical price transmission could roughly offset one another, in which case the estimated elasticity would only coincidentally resemble the elasticity expected under perfect competition. As Weldegebriel (2004, p. 113) summarises: "without knowledge of the functional forms of retail demand and farm input supply, little can be inferred from the numerical value of a price transmission coefficient. In particular, it is not generally possible to attribute low (or high) values of the price transmission coefficient to market power".

10.5. Empirical methods for measuring vertical price adjustment processes

The following section provides a brief overview of the empirical methods used in most studies on vertical (and spatial) price adjustment processes. The workhorse model in these studies is the so-called **error correction model** (ECM), which will be briefly explained below²¹.

The ECM has dominated the empirical literature on price adjustment processes since the mid-1990s.²² Before that, relationships between prices at different stages of the food chain were often analysed with simple correlation coefficients or least-squares regressions. However, these

²¹ The ECM is a key element of cointegration analysis, which is a branch of (time series) econometrics. A comprehensive introduction to cointegration analysis would go beyond the scope of this chapter. Hendry and Juselius (2000, 2001) provide a detailed but accessible introduction for agricultural economists.

²² The first application in a leading agricultural economics journal is Ardeni (1989).

methods were criticised because they were prone to misinterpretation. Prices at different stages of the chain might be correlated because farmers, processors and consumers interact on markets under conditions of more or less perfect competition as described in Gardner's model. In this case, prices at different stages of the chain will be interrelated because the markets in which they are formed are integrated.

But prices can also be correlated because inflation is strong in a country and all nominal prices therefore follow similar rising trends, even when the markets in question are not integrated. Furthermore, prices can be correlated because the state intervenes in price formation by prescribing fixed margins or price formulas (e.g. that the price of flour must be exactly 120% of the price of milling wheat).

Finally, there is also the danger of so-called **spurious correlation**, which can occur when time series variables (such as prices) are not stationary. A stationary time series has a constant mean as well as a constant variance and constant covariances, regardless of which segment of the series is viewed. A stationary time series can temporarily deviate upwards or downwards from its long-term mean, but will always tend back to this mean. Many economic variables such as prices, however, are characterised by non-stationary behaviour; they rise or fall over longer periods and show no inherent tendency to return to a constant mean. In a seminal study, Granger and Newbold (1974) showed that when X_t and Y_t are two independent non-stationary time series (with $t = 1, 2, \dots, T$ observations) and the regression $Y_t = \beta_0 + \beta_1 X_t$ is estimated, the null hypothesis $H_0: \beta_1 = 0$ (which is actually true) is rejected far too often.²³ This means that regressions between non-stationary variables often produce what appear to be statistically significant relationships that are actually just statistical artefacts. Since many prices have non-stationary properties, regressions such as $P_t^A = \beta_0 + \beta_1 P_t^B$ are also susceptible to the phenomenon of spurious correlation.

In summary, a high correlation or significant regression between two prices does not necessarily mean that these prices are actually linked in an

²³ In Granger and Newbold's (1974) experiment with non-stationary variables (so-called random walks), the true null hypothesis $H_0: \beta_1 = 0$ was rejected in about 75% of all cases at the 5% level of significance. In 2003, Clive W.J. Granger, together with Robert F. Engle, received the Nobel Memorial Prize in Economic Sciences in recognition of his many contributions to the analysis of economic time series, including the estimation and interpretation of ECMs.

equilibrium relationship determined by market processes and the behaviour of economic agents. Simple correlation coefficients and least squares estimates are not able to distinguish between price correlations that arise on integrated markets and those that are due to other causes such as inflation or spurious correlation.

As these insights were increasingly recognised, econometricians responded by estimating dynamic regression models with lagged so-called lag terms (e.g. $P_t^A = \beta_0 + \beta_1 P_t^B + \beta_2 P_{t-1}^B + \beta_3 P_{t-1}^A$) or models in differences (e.g. $\Delta P_t^A = \beta_0 + \beta_1 \Delta P_t^B$, where $\Delta X_t = X_t - X_{t-1}$). Then, Engle and Granger (1987) showed how the ECM could be used to estimate relationships between non-stationary time series. A simple ECM for two price series P^R and P^C (R denotes the raw agricultural product and C the consumer product) takes the following form:

$$\Delta P_t^C = \alpha_C (P_{t-1}^C - \beta_0 - \beta_1 P_{t-1}^R) + \delta_{C1} \Delta P_{t-1}^C + \delta_{R1} \Delta P_{t-1}^R \quad (20)$$

The dependent variable on the LHS of equation (20) (ΔP_t^K) measures the change in P^C between the previous and current periods. The term in parentheses on the RHS of equation (20) captures deviations from the long-run equilibrium relationship $P_t^C = \beta_0 + \beta_1 P_t^R$ between P^C and P^R . The non-stationary price series P^C and P^R are connected by a stable long-run equilibrium relationship only if the deviations from this long-run relationship ($P_{t-1}^C - \beta_0 - \beta_1 P_{t-1}^R$, the so-called error correction term) are stationary and thus always tend to return to their mean value of zero. $\beta_1 = \partial P^C / \partial P^R$ is the long-run price adjustment coefficient; if the prices P^R and P^C are in logarithms, β_1 represents the elasticity of vertical price transmission.

If prices are in equilibrium, then $P_t^C = \beta_0 + \beta_1 P_t^R$ and hence the error correction term ($P_t^C - \beta_0 - \beta_1 P_t^R$) equals 0. If the long-run equilibrium is temporarily disturbed, for example by an unexpected shock, then $P_t^C - \beta_0 - \beta_1 P_t^R \neq 0$. The parameter α_C (often called the adjustment parameter) measures how P^C reacts in the next period to correct part of the deviation from equilibrium. For example, if P^C is too large relative to P^R , then a positive deviation or error from the long-run equilibrium exists because $P_t^C - \beta_0 - \beta_1 P_t^R > 0$. If, for example, $\alpha_C = -0.5$ (-0.20), then P^K will fall to correct 50% (20%) of this error in the next period. α_C must be less than zero so that a too high (low) P^C is corrected downwards (upwards) in the next period, thus correcting the error and restoring the long-run equilibrium between P^R and P^C .

The two lagged price change terms on the right-hand side of equation (20), $\delta_{C1}\Delta P_{t-1}^C$ and $\delta_{R1}\Delta P_{t-1}^R$, measure short-term dynamic interactions between prices. Equation (20) contains only one lag each of the price changes (ΔP_{t-1}^C and ΔP_{t-1}^R), but further lags can be added in empirical applications.

A similar ECM can be specified for changes in the raw product price P^R :

$$\Delta P_t^R = \alpha_R(P_{t-1}^C - \beta_0 - \beta_1 P_{t-1}^R) + \gamma_{C1}\Delta P_{t-1}^C + \gamma_{R1}\Delta P_{t-1}^R \quad (21)$$

If P^C is too large relative to P^R ($P_t^C - \beta_0 - \beta_1 P_t^R > 0$), P^R must increase in the next period to contribute to correcting the deviation from long-run equilibrium. Therefore, $\alpha_R > 0$ must hold in equation (21). The sum $|\alpha_C| + \alpha_R$ thus measures how fast the prices P^C and P^R react together to correct temporary deviations from equilibrium. Equations (20) and (21) together form a so-called vector ECM (abbreviated in the literature as VECM for **vector error correction model**). The equations in a VECM are usually estimated simultaneously using a method developed by Johansen (1991).

Case study 4: A VECM for wheat and flour prices in Ukraine

In case study 2, the elasticity of vertical price transmission between wheat and flour prices in Ukraine was estimated using the following least squares regression: $\ln P_t^{\text{flour}} = 1.790 + (0.808 * \ln P_t^{\text{wheat}})$. However, the prices used in this regression might be non-stationary (see Figure 10.4), and if that were true these regression results might reflect spurious correlation. To rule out this possibility, a typical price transmission analysis follows the following steps. First, the individual price series are tested for non-stationarity. So-called unit root tests are used for this purpose. If both series are non-stationary, there is a risk of spurious correlation. Therefore, the second step is to test whether deviations from the estimated long-term relationship between the prices are stationary. So-called **cointegration tests** are used for this purpose. If the deviations are stationary, spurious correlation can be excluded; there is a valid, interpretable long-term relationship between the prices. In this case, third, a VECM can be estimated for these prices.

The technical details of these various test and estimation procedures cannot be explained here. As mentioned above, Hendry and Juselius (2000, 2001) are useful sources. For Ukrainian wheat and flour prices, first, the results of unit root tests show that both prices are non-stationary. Second, cointegration tests show that there is a valid long-run relationship between these prices - the regression results presented above are therefore not due to

spurious correlation. Third and finally, the following VECM can be estimated for the prices (standard errors in brackets):

$$\begin{aligned}\Delta \ln P_t^{\text{flour}} &= -0.124 (\ln P_{t-1}^{\text{flour}} - 1.607 - 0.835 \ln P_{t-1}^{\text{wheat}}) + 0.367 \Delta \ln P_{t-1}^{\text{flour}} + 0.024 \Delta \ln P_{t-1}^{\text{wheat}} \\ &\quad (0.023) \quad (0.246) \quad (0.037) \quad (0.061) \quad (0.049) \\ \Delta \ln P_t^{\text{wheat}} &= 0.002 (\ln P_{t-1}^{\text{flour}} - 1.607 - 0.835 \ln P_{t-1}^{\text{wheat}}) + 0.208 \Delta \ln P_{t-1}^{\text{flour}} + 0.203 \Delta \ln P_{t-1}^{\text{wheat}} \\ &\quad (0.036) \quad (0.246) \quad (0.037) \quad (0.094) \quad (0.074)\end{aligned}$$

The estimated long-run relationship in the VECM $\ln P_t^{\text{flour}} = 1.607 + (0.835 * \ln P_t^{\text{wheat}})$ is slightly different from the long-run relationship presented above in case study 2. The differences arise because the VECM is estimated using the maximum likelihood method rather than least squares. However, the differences are minor and do not change the key results, especially the estimated elasticity of vertical price transmission, which remains slightly above 0.8.

The estimated adjustment parameters $\alpha_{\text{flour}} = -0.124$ and $\alpha_{\text{wheat}} = 0.002$ indicate that the flour price responds when there are deviations from the long-run equilibrium between wheat and flour prices, by about 12.4% of the deviation per week. The wheat price, on the other hand, does not react (the estimated value of 0.2% per week is very small and statistically insignificant). This result is expected because, as explained above, the wheat price in Ukraine depends on the world market, while the flour price is formed on the domestic market. If an imbalance arises between wheat and flour prices in Ukraine, the much larger world market for wheat cannot be expected to react significantly; instead, the comparatively small Ukrainian flour market will adjust to restore equilibrium (the dog wags the tail, not the other way around). Correcting roughly 12.4% of a disequilibrium each week is not very fast; at that rate it will take several months to restore equilibrium. One possible explanation for this result could be that mills do not purchase the wheat they need in small batches on a weekly basis. Instead, they usually purchase larger batches at longer intervals. Hence, the flour a mill sells today was probably milled from wheat purchased several weeks or even months ago. Therefore, it can take time for price signals to be transmitted from wheat to flour markets.

The use of VECMs to estimate price transmission processes has several advantages. First, equilibrium relationships can be distinguished from spurious correlations. Second, not only the long-run equilibrium relationship between two prices is estimated, but also the short-run reactions

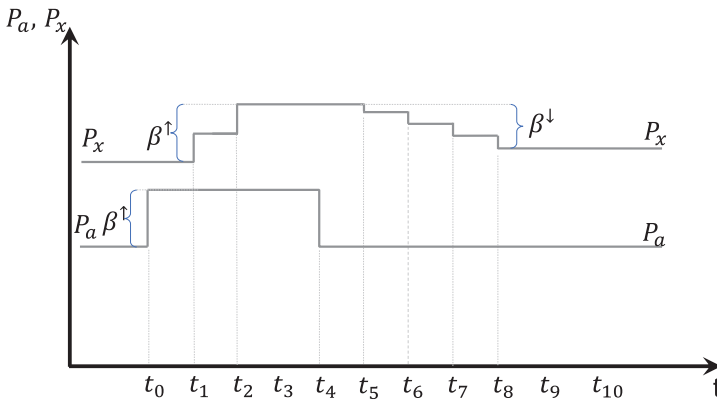
of prices to shocks that disturb this equilibrium. In this way we can gain insights into the pattern and speed of a price adjustment process.

10.6. Asymmetric vertical price transmission

A price adjustment is **asymmetric** if the magnitude and/or speed of the response of one price to a change in another price depends on the direction of the original price change. Figure 10.5 is a schematic depiction of both types of asymmetry: the response of P_x to an increase in P_a is complete after two periods and equals $\beta^\uparrow$, but the response of P_x to a decrease in P_a is only complete after four periods and is smaller ($\beta^\downarrow < \beta^\uparrow$). This type of asymmetry is often called **rockets and feathers**, suggesting that prices rise quickly (like rockets) but fall slowly (like feathers).

Rockets and feathers behaviour is often suspected: car owners and households that heat with oil claim that the prices of gasoline or heating oil rise quickly when the price of crude oil rises, and fall slowly when the price of crude oil falls. Pig farmers sometimes claim that slaughter companies and retailers use their market power to pass price increases for slaughter pigs on to consumers very quickly, because these price increases would otherwise squeeze their margins. However, farmers claim that the same slaughter companies and retailers will pass price reductions for slaughter pigs on to consumers more slowly, because these price reductions stretch their margins, and they are in no hurry to pass on this benefit. Case study 5 shows that empirical evidence of asymmetric vertical price transmission of the rockets-and-feathers type can be found in the German market for slaughter pigs.

Figure 10.5: Schematic depiction of asymmetric vertical price adjustments



Source: Own depiction.

Case study 5: Asymmetric price adjustments in the market for slaughter pigs in Germany

Von Cramon-Taubadel (1998) analyses vertical price transmission between producer prices for slaughter pigs and wholesale prices for pork cuts in northern Germany using weekly data from January 1990 to February 1993. He uses the following asymmetric specification of the ECM:

$$\Delta P_t^{whole} = \alpha_{whole}^+(u_{t-1}^+) + \alpha_{whole}^-(u_{t-1}^-) + \delta_1 \Delta P_{t-1}^{whole} + \gamma_1 \Delta P_{t-1}^{prod} + \gamma_2 \Delta P_{t-2}^{prod} + \gamma_4 \Delta P_{t-4}^{prod} \quad (22)$$

with $u_t = P_t^{whole} - \beta_0 - \beta_1 P_t^{prod}$, the deviation from the long-run equilibrium relationship between the wholesale price of cuts (P^{whole}) and the producer price of slaughter pigs (P^{prod}). Possible asymmetry is captured by decomposing the deviation from equilibrium (u_t) into two components. The component u_t^+ equals the positive values of u_t , and is otherwise zero; the component u_t^- conversely equals the negative values of u_t , and is otherwise zero:

$$u_t^+ = \begin{cases} u_t & \forall u_t > 0 \\ 0 & \forall u_t \leq 0 \end{cases} \text{ and } u_t^- = \begin{cases} u_t & \forall u_t < 0 \\ 0 & \forall u_t \geq 0 \end{cases}$$

The corresponding symmetric ECM

$$\Delta P_t^{whole} = \alpha_{whole}(u_t) + \delta_1 \Delta P_{t-1}^{whole} + \gamma_1 \Delta P_{t-1}^{prod} + \gamma_2 \Delta P_{t-2}^{prod} + \gamma_4 \Delta P_{t-4}^{prod} \quad (23)$$

is a restricted form of the asymmetric ECM in equation (22), in which the restriction $\alpha_{whole}^+ = \alpha_{whole}^-$ is imposed (since $u_t^+ + u_t^- = u_t$). The corresponding null hypothesis $H_0: \alpha_{whole}^+ = \alpha_{whole}^-$ can therefore be tested by means of an F-test. For the German pork price data this H_0 can be rejected (p-value equals roughly 0.02). The following (simplified) estimation results for the asymmetric ECM are presented (see von Cramon-Taubadel 1998, Table 2; standard errors in brackets):

$$\Delta P_t^{whole} = -0.057(u_{t-1}^+) - 0.270(u_{t-1}^-) + 0.164\Delta P_{t-1}^{whole} + 0.114\Delta P_{t-1}^{prod} - 0.093\Delta P_{t-2}^{prod} - 0.107\Delta P_{t-4}^{prod}$$

(0.041) (0.083) (0.062) (0.054) (0.045) (0.035)

These results indicate a rockets and feathers asymmetry. $u_t > 0$ implies that the wholesale price P^{whole} is too high relative to the producer price P^{prod} , i.e. the slaughter companies are enjoying an above-average margin. If this is the case, only a relatively small and statistically insignificant correction ($\alpha_{whole}^+ = -0.057$) takes place. $u_t < 0$ implies that the wholesale price P^{whole} is too low relative to the producer price P^{prod} , i.e. that the slaughter margin is below-average. In this case, the correction is about five times larger ($\alpha_{whole}^- = -0.270$). Deviations from the long-term equilibrium that stretch the margin are thus corrected significantly more slowly than those that squeeze the margin. Simulations for Germany as a whole, calculated under the assumption of symmetric price adjustment with an adjustment parameter of $\alpha_{whole}^+ = \alpha_{whole}^- = 0.270$, show that asymmetry increased the slaughter margin by about 28 million euros per year.

However, results such as those presented in case study 5 do not conclusively prove that the companies in the German slaughter industry have used market power to expand their margins and thus their profits during the observation period. In terms of the classical **structure-conduct-performance** approach (SCP, see section 7 in Chapter 8), we observe a certain market *structure* (many pig farmers, but few large slaughter companies) and a certain market *outcome* (asymmetric vertical price adjustments). But since we do not directly observe the *behaviour* (conduct) of the slaughter companies, we cannot be sure that the market structure is causing the outcome. The fact that asymmetric price transmission is observed does not prove that the companies colluded or otherwise coordinated their behaviour to generate it. It would therefore be premature to attribute the observed asymmetry to market power. This caution is called for because, first, **asymmetric price adjustment** can be caused by factors other than the exercise of market power and, second, because market power does not necessarily lead to rockets and feathers asymmetry.

10.6.1. Alternative causes of asymmetric price adjustment

Asymmetric price adjustments can occur in the absence of market power. Several alternative causes of asymmetry have been proposed in the literature.²⁴ According to Kinnucan and Forker (1987), one possible explanation for asymmetric price adjustments could be agricultural policy. If farmers in a country traditionally benefit from price support, processors will expect the government to step in and support farmers' prices whenever they fall. Processors will therefore expect farm-gate price reductions to be short-lived. Accordingly, they will not reduce the sales prices of their consumer products when farm-gate prices fall. In comparison, processors will expect increases in farm-gate price to be longer-lived, and they will therefore be more likely to pass on the resulting cost increases to their customers. The result will be rockets-and-feathers price adjustment that is not caused by market power.

Another alternative explanation for asymmetric price adjustments is proposed by Ball and Mankiw (1994). In their model, inflation in conjunction with so-called **menu costs** causes asymmetric price adjustment. Menu costs are the costs incurred by a firm when it changes its prices. In the original sense of the words, menu costs refer to the costs that a restaurant incurs when it adjusts its prices and therefore has to redesign and reprint its menu. More generally, the term refers to all costs associated with changing price lists, catalogues, websites, etc. As a result of menu costs, a firm will not adjust its sales prices in response to every small change in its input costs. Instead, it will only adjust its sales prices if the profit loss due to the input cost change exceeds the menu costs caused by the sales price change. Some firms will only adjust their sales prices at regular, often seasonal intervals (e.g. the summer catalogue, the autumn collection, the Christmas menu, etc.), and use these scheduled opportunities to respond to any input price changes that have occurred in the interim.²⁵

²⁴ Surveys of the literature on asymmetric price adjustment processes are provided by Frey and Manera (2007) and Meyer and von Cramon-Taubadel (2004). Loy et al. (2016) provide a comprehensive recent overview of the various causes of asymmetric vertical price adjustment that have been proposed in the literature.

²⁵ Firms might also worry that too frequent adjustments of their sales prices will confuse or annoy customers. Rotemberg (2010) argues that this danger is particularly present in the case of storable products: if prices increase for a storable product, consumers will regret that they did not buy it earlier; if prices fall, consumers will regret that they did not wait longer. This worry on the part of firms will, like menu costs, make them less likely to change their sales prices.

In Ball and Mankiw's model, firms must take both menu costs and inflation into account when setting prices. All other things being equal, over time inflation causes a firm's margin to decline in real terms. After a certain period of time, its margin will have fallen so much that a firm will respond by increasing its sales price, despite menu costs. If, however, input prices have fallen in the meantime, this counteracts inflation; it increases the firm's margin and enables the firm to postpone or cancel the sales price increase (and save the menu costs that it would otherwise have incurred). If, instead, input prices have increased in the meantime, the firm's margin is doubly squeezed (by both inflation and the input price increase). In this case the firm will be obliged to increase its sales price sooner. Viewed from outside, it will appear that input price reductions do not or less often lead to adjustments in sales prices, compared with input price increases. The result, again, is asymmetric price adjustment that is not caused by the exercise of market power.

Yet another explanation for asymmetric price adjustment proposed in the literature relates to so-called **stock-out costs**. A stock-out occurs when a product that is normally part of a retailer's assortment is sold out and therefore temporarily unavailable to customers. If a stock-out occurs, the retailer incurs costs in the form of reduced sales. However, there is also the danger that customers will look for and find the missing product at a competing retailer and, in the worst case, switch to the competing retailer altogether. As Gaur and Park (2007, p. 227) state: "in practice, customers do react substantially and negatively to poor service (e.g., stock-outs), which may lead them to switch retailers on subsequent trips". This risk is particularly high for perishable staples such as fresh milk, which households cannot store in large quantities; if such a product is stocked-out in one store, customers are very likely to look for it elsewhere.

To reduce the risk of stock-outs, retailers might therefore tend to reduce their sales prices only slowly when their input prices fall, especially for perishable products, so as not to stimulate demand too much. If, on the other hand, purchase prices rise, retailers will pass this cost increase on to their customers more quickly. This asymmetry will not occur or will be less pronounced for products that are easier and less expensive to stock.²⁶

²⁶ Slonim and Gabarino (2009) develop a formal model in which the strength of the resulting asymmetry in vertical price adjustment depends, among other things, on the storability/perishability of the product.

Finally, an explanation for asymmetric price adjustments can also be derived from the results of Gardner's model, which was presented above. In section 10.3.3 we saw that the elasticity of vertical price transmission is higher when price adjustments are triggered by a change in the price of the consumer product x , rather than by a change in the price of the raw farm product a ($E_{P_x, P_a} < E_{P_x, P_a}$). If, over time, price reductions are more often triggered by shocks on the market for a , while price increases predominantly arise as a result of shocks on the market for x , rockets and feathers asymmetry can occur even under conditions of full competition.

10.6.2. Does market power necessarily lead to rockets and feathers asymmetry?

It certainly is conceivable that firms that have oligopoly power might collude to coordinate rockets and feathers asymmetry. Over time, this would increase average margins for these firms. Of course, the firms could simply agree to permanently raise their sales prices in order to permanently increase their margins. However, if they worry that permanent increases might quickly come to the attention of competition authorities, they might consider rockets and feathers behaviour as a subtler and less conspicuous way of turning market power into profits.

However, collusive price manipulation is not the only behaviour that can arise in concentrated markets. Imagine, for example, an oligopoly consisting of a financially strong firm that has substantial financial reserves, and several other firms that are in financially weaker positions and, in some cases, highly indebted. In such a situation, the strong firm might be willing to forego profits in the short run in order to drive some competitors out of the market and thus expand its market share in the medium or long run. If the firm pursues such a strategy, it is unlikely to participate in collusion that leads to rockets and feathers asymmetry. After all, rockets and feathers asymmetry would increase average margins over time, and thus reduce the pressure on financially weaker firms. Instead, the strong company might try to implement the opposite asymmetry. If input prices were to fall, the strong firm would immediately reduce its sales prices to maintain economic pressure on its competitors. If input prices were to rise, the strong firm could leave its sales prices unchanged and temporarily absorb below-average margins in the hope that these might force some weaker competitors to exit the market. The result would be the exact opposite of rockets and feathers.

In conclusion, asymmetric price adjustments do not necessarily result from the exercise of market power, and the exercise of market power does not necessarily lead to rockets and feathers asymmetry. The following case study presents evidence for the dairy market in Germany.

Case study 6: Price transmission and market power in the German dairy market

Loy et al. (2016) analyse an extensive dataset of weekly price data for 90 different brands of fluid milk (3.5% fat content in one litre containers) in 327 retail stores from 2005 to 2008. The retail price of milk is strongly influenced by the wholesale prices that dairies charge. The authors analyse which factors, such as market power, influence the transmission of prices between the wholesale and the retail level in the 327 shops.

In a first stage, the authors estimate an ECM for each brand of milk in each shop (referred to as "product" in the following). Both symmetric (α) and asymmetric adjustment parameters (α^+ and α^-) are estimated (see case study 5). These indicate how quickly prices for a particular product react to deviations from the long-run equilibrium and whether these price reactions are symmetric. In a second step, the authors use the adjustment parameters estimated in the first stage as endogenous variables in cross-section regressions. The aim of these cross-section regressions is to test whether there is a systematic relationship between the speed of vertical price adjustment or the extent of estimated asymmetry on the one hand, and metrics of market power and other characteristics of dairy markets on the other.

The results reveal a negative correlation between a retailer's market power (measured using the price-cost margin, see chapter 8) and the speed of price adjustment. In other words, retailers with more market power appear to adjust their retail prices more slowly in response to wholesale prices changes. The authors also find clear evidence of rockets and feathers asymmetry. However, they find that the extent of this asymmetry decreases with increasing market power. This result contradicts the common expectation that rockets and feathers asymmetry is a consequence of market power. Finally, the authors find that price adjustment asymmetry is significantly stronger for fresh milk than for ultra-heat treatment (UHT) milk that can be stored for longer periods of time. They interpret this result as a possible indication that asymmetric price adjustment is being caused by retailers' reactions to stock-out costs.

10.7. Implications for agricultural and food policy

As noted in the introduction to this chapter, vertical pricing is a controversial topic. Farmers and other stakeholders often call for political interventions such as minimum farm-gate prices, bans on the sale of food below production costs, and restrictions on market power in food processing and retailing. What conclusions for agricultural and food policy can we draw based on the economic concepts and models presented in this chapter?

Competition policy

First, we have seen that the effects of market power on vertical price transmission are complex. The exercise of market power can increase but also reduce elasticities of vertical price transmission. It can lead to asymmetric price adjustment, but asymmetry can also have other causes. The empirical analysis of prices at different levels of the food chain, e.g. by means of ECMs, can provide important insights into price adjustment processes, but can only generate limited evidence on the existence and exercise of market power. Therefore, and as explained in detail in chapter 8, competition policy must also rely on other analytical methods and instruments. In this chapter we have focussed on vertical price relationships in the food chain. However, market participants negotiate not only prices, but also other contractual conditions such as supply guarantees, exclusivity, payment conditions, and premiums and penalties for quality, punctuality, and other contract fulfilment criteria. Thus, analysis of price data alone can only be one ingredient in a comprehensive competition policy.

Food processing and food retailing in Germany are characterised by high concentration. Concentration does not automatically imply the exercise of market power. However, in recent decades the Germany Federal Cartel Office has investigated many branches of the food processing industry as well as food retailing, and these investigations have often led to the imposition of fines. The situation is similar in many other industrialised countries. Accordingly, there are indications that competition is less than perfect in the food chain in many countries. This underlines the crucial need for competition authorities that are equipped with sufficient resources and regulatory power.

Minimum prices and other interventions in price formation

As explained in section 10.2, the widening of the market margin for food over time is due to many factors including technical change, rising incomes, and changes in farm and consumer households' preferences and behaviour. The widening of the market margin therefore does not in itself provide a justification for government intervention.

In public debates it is often argued that food prices in Germany are too low and that policy makers should intervene to correct or improve price formation in the food chain. Statements such as the following quote by the President of the German Farmers' Union Joachim Rukwied meet with approval among the population: "Food should not be dumped at bargain-basement prices" (DBV 2020). In a survey conducted in early 2020 (Heinrich 2020), 73% of respondents were in favour of a ban on the sale of food at prices below production costs.

The implementation of such a ban, however, would be associated with considerable difficulties and side effects. First, references to "the" production costs fail to recognise that agriculture is heterogeneous. The production costs of a litre of milk or a ton of wheat can vary, both regionally and from farm to farm. If average production costs were used as the minimum price in food retailing, food would continue to be sold at prices below the production costs of farms that have above-average costs. If, instead, the production costs of the marginal supplier were used to determine the minimum price, very high margins would arise in the food chain.

Second, the enforcement of a minimum retail price would increase consumer spending on food and retailers' revenues, but it would not necessarily lead to higher prices for farmers or a larger farm share in consumer expenditure on food. For the latter to happen, the state would have to regulate not only retail prices but also pricing along the entire food chain. In any event, it cannot be assumed that the percentage increase in farm-gate prices would equal the percentage increase in consumer prices. This is because, as shown above, the elasticity of vertical price transmission is only equal to one under very restrictive conditions.

Third, while it is conceivable that minimum prices could be set for comparatively simple food products such as milk, eggs and flour, this would be much more complicated for highly processed food products that are composed of several raw products such as frozen pizzas and muesli mixes.

In order to ensure that the cost of production (however defined) of each individual ingredient is not undercut, policy makers would require extensive and constantly updated information on prices, recipes and the costs of complementary goods and services. It is unlikely that the state would be able to obtain this information and use it effectively. In the planned economies of Eastern Europe before 1989, policy makers attempted to control prices and margins along the entire food chain; the results were chronic shortages, often very limited product variety, and considerable inefficiencies in the system.

Finally, calls for minimum prices and bans on dumping in the food chain often implicitly assume that Germany (or whatever other country is under consideration) is a closed economy, i.e. they fail to account for international trade. In the single European market, not everything that is sold below production costs in a particular country is automatically a case of dumping. For example, it is not dumping if a processor combines expensive ingredients from Germany with less expensive ingredients from other EU countries, and sells the end product at a price that reflects the average ingredient costs. Similarly, it is not dumping if a processing firm sells a product in Germany that was produced exclusively on the basis of raw materials from other EU countries where production costs are lower than in Germany.²⁷

If there is imperfect competition in the food chain, it should be addressed using the instruments of competition policy (see chapter 8); minimum prices and other market interventions will generally be less effective and less efficient. In some cases, legislation and government programmes can help farmers coordinate and bundle their market power and, thus, better assert their interests vis-à-vis other stakeholders in the food chain. From the 1960s until the 1990s the EU provided farmers with high levels of price support. Nevertheless, the farm share in consumer expenditure on food fell and structural change in agriculture continued. There is no reason to expect that these trends could be arrested by re-implementing price support today.

²⁷ Provided, of course, that the imported raw materials were produced in compliance with the quality standards (health, animal welfare, environment, etc.) that apply in the EU, and that were adopted by the EU with Germany's participation.

10.8. Keywords

- Agricultural terms of trade
- Asymmetric price adjustment
- Complementary goods and services
- Convenience foods
- Cointegration test
- Elasticity of factor substitution
- Elasticity of vertical price transmission
- Equilibrium displacement form of an equation
- Error correction model (ECM)
- Market margin
- Menu costs
- Price vector
- Quantity vector
- Rockets and feathers
- Spurious correlation
- Stock-out costs
- Structure, conduct, performance (SCP)
- Unit margin
- Transformation coefficient
- Vector error correction model (VECM)
- Vertical price relationships

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Appendix 10.1: The solution of Gardner's model

The six initial equations of Gardner's (1975) model are:

$$x = f(a, b) \quad (\text{supply of the consumption product } x, \text{ a production function}) \quad (4)$$

$$x = g(P_x) \quad (\text{demand for the consumption product } x) \quad (6)$$

$$a = h(P_a) \quad (\text{supply of the raw agricultural product } a) \quad (7)$$

$$\frac{\partial(f(a, b))}{\partial a} P_x = P_a \quad (\text{demand for the raw agricultural product } a) \quad (8)$$

$$b = k(P_b) \quad (\text{supply of the non-agricultural input } b) \quad (9)$$

$$\frac{\partial(f(a, b))}{\partial b} P_x = P_b \quad (\text{demand for the non-agricultural input } b) \quad (10)$$

These equations are put into *equilibrium displacement* form by totally differentiating them and converting all unknowns into proportional changes. For equation (4), this leads to the following steps, where a variable with a dot (e.g. $\dot{x}$) represents the proportional change of this variable ($\dot{x} = dx/x$):

$$dx = \frac{\partial(f(a, b))}{\partial a} da + \frac{\partial(f(a, b))}{\partial b} db \quad (\text{totally differentiate equation 4})$$

$$\frac{dx}{x} = \frac{\partial(f(a, b))}{\partial a} \frac{a}{x} \frac{da}{a} + \frac{\partial(f(a, b))}{\partial b} \frac{b}{x} \frac{db}{b} \quad (\text{divide by } x, \text{ multiply RHS terms by } \frac{a}{a} \text{ and } \frac{b}{b}, \text{ respectively})$$

$$\frac{dx}{x} = \frac{P_a}{P_x} \frac{a}{x} \frac{da}{a} + \frac{P_b}{P_x} \frac{b}{x} \frac{db}{b} \quad (\text{replace } \frac{\partial(f(a, b))}{\partial i} \text{ using equations 8 and 10})$$

$$\frac{dx}{x} = S_a \frac{da}{a} + S_b \frac{db}{b} \quad (S_i = \frac{iP_i}{xP_x}, \text{ the cost share of production factor } i)$$

$$\dot{x} = S_a \dot{a} + S_b \dot{b} \qquad \frac{dx}{x} = \dot{x}; \frac{da}{xa} = \dot{a}; \frac{db}{b} = \dot{b} \quad (4')$$

Equations (6), (7) and (9) all have the same form and are linearised as follows:

$$m = l(P_m) \qquad \text{where } m = \{x, a, b\} \text{ and } l = \{g, h, k\}, \text{ respectively.}$$

$$dm = \frac{\partial(l(P_m))}{\partial P_m} dP_m \qquad (\text{totally differentiate})$$

$$\frac{dm}{m} = \frac{\partial(l(P_m))}{\partial P_m} \frac{P_m}{m} \frac{dP_m}{P_m} \qquad (\text{divide by } m, \text{ multiply RHS by } \frac{P_m}{P_m})$$

$$\dot{m} = \epsilon_{m,P_m} \dot{P}_m \qquad (\epsilon_{m,P_m} = \frac{\partial(l(P_m))}{\partial P_m} \frac{P_m}{m}, \text{ the own-price elasticity of demand for } m = x, \text{ or supply for } m = a \text{ or } b).$$

These derivations therefore lead to:

$$\dot{x} = \eta_{x,P_x} \dot{P}_x \qquad (\eta_{x,P_x} \text{ is the own-price elasticity of demand for } x) \quad (6')$$

$$\dot{a} = \epsilon_{a,P_a} \dot{P}_a \qquad (\epsilon_{a,P_a} \text{ is the own-price elasticity of supply of } a) \quad (7')$$

$$\dot{b} = \epsilon_{b,P_b} \dot{P}_b \qquad (\epsilon_{b,P_b} \text{ is the own-price elasticity of the supply of } b) \quad (9')$$

The linearisation of equations (8) and (10) is somewhat more complicated, as it requires derivatives of the marginal products $\frac{\partial(f(a,b))}{\partial a}$ and $\frac{\partial(f(a,b))}{\partial b}$ (i.e. second derivatives of the production function $x = f(a, b)$). The following notation simplifies the presentation: $\frac{\partial(f(a,b))}{\partial a} = f_a$, and $\frac{\partial(f(a,b))}{\partial b} = f_b$. Equation (8) is transformed as follows:

$$P_a = f_a P_x$$

$$dP_a = \frac{\partial P_a}{\partial P_x} dP_x + \frac{\partial P_a}{\partial f_a} df_a \qquad (\text{totally differentiate})$$

$$dP_a = f_a dP_x + P_x df_a \quad \left(\frac{\partial P_a}{\partial P_x} = f_a, \text{ and } \frac{\partial P_a}{\partial f_a} = P_x \right)$$

$$dP_a = f_a dP_x + P_x \left(\frac{\partial f_a}{\partial a} da + \frac{\partial f_a}{\partial b} db \right) \quad (\text{expand } df_a)$$

$$\frac{dP_a}{P_a} = f_a \frac{dP_x}{P_a} + \frac{P_x}{P_a} (f_{aa} da + f_{ab} db) \quad (\text{divide by } P_a, \frac{\partial f_a}{\partial a} = f_{aa}, \frac{\partial f_a}{\partial b} = f_{ab})$$

$$\frac{dP_a}{P_a} = \frac{P_a}{P_x} \frac{dP_x}{P_a} + \frac{P_x}{P_a} (f_{aa} da + f_{ab} db) \quad (f_a = \frac{P_a}{P_x} \text{ from equation 8})$$

We now use the following definitions of the elasticity of factor substitution according to Allen (1938, p. 343): $\sigma = -\frac{b}{a} \frac{f_a f_b}{x f_{aa}}$ and $\sigma = \frac{f_a f_b}{x f_{ab}}$, or equivalently $f_{aa} = -\frac{b}{a} \frac{f_a f_b}{\sigma x}$ and $f_{ab} = \frac{f_a f_b}{\sigma x}$.

$$\dot{P}_a = \frac{P_a}{P_a} \frac{dP_x}{P_x} + \frac{P_x}{P_a} \left(-\frac{b}{a} \frac{f_a f_b}{\sigma x} da + \frac{f_a f_b}{\sigma x} db \right)$$

$$\dot{P}_a = \dot{P}_x - \frac{P_x}{P_a} \frac{b}{a} \frac{P_a P_b}{P_x P_x} \frac{1}{\sigma x} da + \frac{P_x}{P_a} \frac{P_a P_b}{P_x P_x} \frac{1}{\sigma x} db \quad \left(f_a = \frac{P_a}{P_x} \text{ and } f_b = \frac{P_b}{P_x} \text{ from equations 8 and 10} \right)$$

$$\dot{P}_a = \dot{P}_x - \frac{P_x}{P_a} \frac{b}{a} \frac{P_a P_b}{P_x P_x} \frac{1}{\sigma x} da + \frac{P_x}{P_a} \frac{P_a P_b}{P_x P_x} \frac{1}{\sigma x} \frac{b}{b} db \quad (\text{multiply the last term on the RHS by } \frac{b}{b})$$

$$\dot{P}_a = \dot{P}_x - \frac{b P_b}{x P_x} \frac{1}{\sigma} \frac{da}{a} + \frac{b P_b}{x P_x} \frac{1}{\sigma} \frac{db}{b} \quad (\text{simplify})$$

$$\dot{P}_a = \dot{P}_x - \frac{S_b}{\sigma} \dot{a} + \frac{S_b}{\sigma} \dot{b} \quad \left(\frac{b P_b}{x P_x} = S_b \right) \quad (8')$$

Analogously, equation (10) can be transformed into

$$\dot{P}_b = \dot{P}_x + \frac{S_a}{\sigma} \dot{a} - \frac{S_a}{\sigma} \dot{b} \quad (10')$$

The model's six equations in *equilibrium displacement* form are thus summarised:

$$\dot{x} = S_a \dot{a} + S_b \dot{b} \quad (4')$$

$$\dot{x} = \eta_{x, P_x} \dot{P}_x \quad (6')$$

$$\dot{a} = \varepsilon_{a, P_a} \dot{P}_a \quad (7')$$

$$\dot{P}_a = \dot{P}_x - \frac{S_b}{\sigma} \dot{a} + \frac{S_a}{\sigma} \dot{b} \quad (8')$$

$$\dot{b} = \varepsilon_{b, P_b} \dot{P}_b \quad (9')$$

$$\dot{P}_b = \dot{P}_x + \frac{S_a}{\sigma} \dot{a} - \frac{S_b}{\sigma} \dot{b} \quad (10')$$

In matrix form this system of six equations and six unknowns is:

$$\begin{bmatrix} 1 & 0 & -S_a & 0 & -S_b & 0 \\ 1 & -\eta_{x, P_x} & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -\varepsilon_{a, P_a} & 0 & 0 \\ 0 & -1 & S_b/\sigma & 1 & -S_b/\sigma & 0 \\ 0 & 0 & 0 & 0 & 1 & -\varepsilon_{b, P_b} \\ 0 & -1 & -S_a/\sigma & 0 & S_a/\sigma & 1 \end{bmatrix} * \begin{bmatrix} \dot{x} \\ \dot{P}_x \\ \dot{a} \\ \dot{P}_a \\ \dot{b} \\ \dot{P}_b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Exercises

1. Determinants of the marketing margin.
 - a) Discuss the impact of future technological developments (e.g. digitalisation) and new regulations (e.g. labelling requirements for food) on the farm share in consumer expenditure on food.
 - b) The published data on the farm share in consumer expenditure on food in Germany do not distinguish between raw products from conventional and organic agriculture. For which type of agriculture do you expect that the farm share of consumer expenditure will be higher? Justify your answer.
2. The following error correction model was estimated using weekly producer prices for slaughter pigs (P^{prod}) and wholesale prices for pork cuts in northern Germany (P^{whole}) in the period January 1990 and February 1993 (standard errors in parentheses):

$$\Delta \ln P_t^{whole} = -0.100 (\ln P_{t-1}^{whole} - 0.746 - 0.676 \ln P_{t-1}^{prod}) + 0.006 \Delta \ln P_{t-1}^{whole} + 0.249 \Delta \ln P_{t-1}^{prod}$$

(0.045) (0.047) (0.043) (0.093) (0.058)

$$\Delta \ln P_t^{prod} = 0.016 (\ln P_{t-1}^{whole} - 0.746 - 0.676 \ln P_{t-1}^{prod}) - 0.166 \Delta \ln P_{t-1}^{whole} + 0.327 \Delta \ln P_{t-1}^{prod}$$

(0.086) (0.047) (0.043) (0.172) (0.110)

- a) What is the estimated elasticity of transmission between producer and wholesale prices?
 - b) According to the estimates, which of the two prices responds significantly to deviations from the long-run equilibrium between producer and wholesale prices?
 - c) Assume that the prices are initially in equilibrium and that the producer price is 2.00 euro/kg. An outbreak of African swine fever in Poland leads to a 0.25 euro/kg increase in producer prices in Germany. Describe the adjustment paths of the two prices. Roughly how many weeks will it take for the two prices to return to equilibrium?
3. Political intervention.
- a) On 4 June 2009, the Parliament of Ukraine passed Law No. 1447-VI "On amendments to certain laws of Ukraine respecting the improvement of state regulation of the agricultural market". One clause of the law stipulates that the market margin in the food chain shall not exceed 20%. Do you think that this clause can be implemented, and what effects might it have on the agri-food sector in Ukraine?
 - b) Nivievskyi and von Cramon-Taubadel (2011) report that local administrations in Belarus force farms that produce milk in their respective regions to deliver all their milk to specific local dairies, even if other dairies in neighbouring regions offer higher prices for the milk. What effects would you expect from this regulation, for the milk farms but also for the development of the dairy sector as a whole?